

Data request — Collide "Dynamometer Discussions" well

What we would need to take this from "why the undulations?" (answered) to a full pump diagnosis. Ordered by how much each item actually changes the answer, not by how easy it is to ask for.

Everything already posted in the thread is listed at the bottom under [Already have](#) — no need to resend those.

Already answered without any new data

These follow from stroke, depth, speed and the card alone, and none of them move if the items below change:

Result	Value
Rod-string natural frequency N_o	58.3 SPM
Load-peak interval $dt = 60/N_o'$	1.03 s
N/N_o'	0.110
Predicted undulations per half-stroke	4.56 (4 clear humps observed)
String spring rate K_r	271.7 lb/in
S_{kr}	11,141 lb
Rod weight in air W_r	6,863 lb
PPRL / MPRL	12,438 / 9,274 lb
Card area	5,829 ft-lb/stroke
Polished-rod power	1.13 hp

The undulations are normal. At $N/N_o' = 0.11$ a 4,200 ft string will always ring roughly four times per half-stroke; the absence of undulations is what would be surprising.

Tier 1 — blocking. Without these, no diagnosis is defensible.

1. Rod string, as actually run — ANSWERED

"4200' of 3/4" steel rods" — Reed, Collide thread, 27 July 2026

Single-diameter, so the frequency taper factor $F_c = 1.000$ and every timing result above stands as computed.

This answer sharpens the next question rather than closing the matter. The measured minimum polished-rod load (9,274 lb) is **2,412 lb heavier than the entire rod string weighs in air** (6,863 lb, being 4,200 ft ×

1.634 lb/ft). That is not mechanically possible: friction acts upward on the downstroke, so it lowers polished-rod load and widens a card — it cannot lift the whole card.

Two explanations were open. With the string confirmed as recorded, only one survives: **the load cell**.

2. Load-cell make, model, and last calibration date — now the live question

If the cell carries a zero or scale offset, the whole card shifts and the correction is arithmetic. Worth checking whether it was zeroed with the rods hanging rather than unloaded.

Until this is settled, every absolute load number from this card is suspect. Load *differences* — and therefore the undulation analysis above — remain sound, because a fixed offset cancels out of a difference.

3. Runtime — hours per day the unit actually pumped

Was the unit running 24 h, or cycling on a pump-off controller / timer?

Why it matters more than it looks: a unit on a 50% duty cycle produces half the fluid with a perfectly healthy pump. That is indistinguishable from 50% fillage if you only look at daily volume. Any efficiency number we quote without runtime is not a measurement, it's an assumption — so we report it as undetermined rather than guess.

Tier 2 — unlocks the calculated downhole card

This is the piece both commenters in the thread asked for, and the only way to separate pump-off from gas interference from a worn pump.

4. Tubing ID and anchor status

Tubing size (2-3/8 in, 2-7/8 in ...) and whether the tubing is **anchored**. Unanchored tubing breathes with the rod string and changes both the card shape and the effective plunger stroke.

5. Fluid level above the pump

From an acoustic shot if one exists. We currently assume the pump is submerged to its setting depth, which is the pessimistic bound.

6. Oil gravity (°API) and formation volume factor **Bo**

Needed to convert surface barrels to reservoir barrels before any efficiency claim.

7. Produced-fluid viscosity, or bottomhole temperature

Sets rod-string damping, which controls how much of the ringing survives down to the pump.

Note on precision: items 5 and 6 matter less than they appear. Sweeping specific gravity 0.80–0.90 and fluid level 3,000–4,300 ft moves theoretical displacement only from **38.3 to 41.7 bfpd** — about ±4%. Reasonable estimates are fine; we do not need exact numbers. Runtime (item 3) swings the answer far harder than either.

Tier 3 — unlocks the gearbox / torque / power checks

8. Pumping unit API designation

The full nameplate, e.g. C-228D-200-74 . The C-66 mentioned in the thread is an **Arrow Engine natural-gas engine** — the prime mover — not a pumping-unit or gearbox rating, so it doesn't give us a torque limit.

9. Unit geometry dimensions and counterbalance

The API 11E dimension sheet for that unit (A , C , I , K , P , crank radius, phase angle), plus counterbalance moment and structural imbalance. Without these the torque calculation cannot run at all.

10. Motor nameplate

Horsepower, voltage, NEMA class — for the electrical/power side.

Tier 4 — better data beats more data

11. The raw card file, not a screenshot

A .dyn export or a timestamped CSV of load and position, straight from the controller.

Why this is worth more than several items above: time read off a position-axis card is inherently uncertain, because polished-rod velocity goes to zero at both stroke ends. The same ± 1.5 in digitizing error costs:

Peak position	Rod velocity	Time uncertainty
7.5 in	10.62 in/s	± 0.14 s
23.5 in	13.59 in/s	± 0.11 s
35.0 in	9.71 in/s	± 0.15 s

With uncertainties that size, the apparent peak-to-peak spacings of 1.24 s and 0.95 s are **not distinguishable from each other**, and both are consistent with the predicted 1.03 s. A raw time-series file removes this entirely.

12. A downhole card from the controller, if it can produce one

Lets us check our calculated card against the controller's, which is the fastest possible validation of both.

13. Several consecutive strokes, and a card from a different day

The posted card shows overlaid strokes ("Display all"). Stroke-to-stroke variability distinguishes a steady condition from an intermittent one, and a second date shows whether anything is trending.

Already have

From the thread, no need to resend:

- Surface stroke 41 in · pump setting depth 4,300 ft · 6.4 SPM
- 1-1/4 in top-hold-down insert pump

- Tubing pressure building to 150 psi (Baird valve setpoint) · casing 25 psi
- Production 23 bbl oil, 0 water, previous day
- The surface card itself (digitized to ± 40 lb / ± 0.3 in)

What we'd return

With Tier 1 alone: a defensible statement on whether the card is self-consistent, and an honest production check.

With Tier 1 + 2: the calculated downhole pump card, pump fillage, and a diagnosis separating pump-off, gas interference and mechanical wear — the question the thread actually ended on.

With Tier 3: gearbox torque loading against the unit's rating, counterbalance condition, and power consumption.

Beyond this well — building a calibration set

This section is general. Forward it to anyone it might suit.

Where the work actually stands

The physics is done. Surface-to-downhole card conversion reproduces reference cards to within about 1% (median 0.9% normalised RMSE, correlation 1.000 across five validation wells), including deviated hole.

The **diagnostic layer is not**. Pattern recognition is currently calibrated on synthetically generated cards, meaning it recognises shapes a computer drew rather than shapes a well made. On a real field card it returns answers that do not survive scrutiny — including a rod-parting call, at full confidence, on a well that produced 23 barrels the day before.

That is a data problem, not an algorithm problem, and it is stated here rather than papered over.

What is actually valuable

A dynamometer card on its own contributes almost nothing to calibration. What makes one valuable is **knowing how it turned out**. The outcome is the label, and without labels more cards teach a classifier nothing.

In rough order of usefulness:

Value	What	Why
Highest	Card + what was found on the pull	The workover or pulling report is the important half
Highest	Before/after pair on one well	Same string, known change — the cleanest possible signal
High	Confirmed healthy cards	Harder to come by than sick ones, and the baseline everything is measured against
Useful	Raw <code>.dyn</code> or timestamped CSV	Removes the timing uncertainty that screenshots carry
Useful	Repeat cards over months	Shows what a condition looks like developing, not just arrived

Minimum context to make a card usable: rod string, pump size and setting depth, stroke, SPM, and roughly what the well makes.

The trade, plainly

You send data. We do the analysis and send it back — your wells, worked properly, at no charge. That part starts immediately, not in three months.

This is an open source effort, not a product built toward a licence fee. The code is already public. Check the arithmetic, hand it to your own engineer, or ignore us entirely and still use it. There is nothing to get locked into, and nothing that can be taken away later.

Early contributors are supported for life. As stated — not an introductory rate that changes once the thing works. The people who contribute cards while the diagnostics are still rough are the reason it becomes trustworthy at all, and that earns permanent standing.

It does not stop at dynacards. Rod string design, unit sizing, gas interference, production troubleshooting — anything across your operations. Better to be useful broadly than narrowly correct about one card.

Alongside that:

- A diagnostics package: surface card in, downhole card and a ranked list of likely conditions out, with a **measured** accuracy figure behind it rather than a marketing one
- Your own wells run through it first
- Well identifiers kept out of anything public — the existing validation set is anonymised the same way

Target is roughly three months from a workable set of labelled cards.

The honest caveat, kept in view

Until a real-card accuracy number exists, any diagnosis from this system should be read as a hypothesis to check against the well, not a finding. The calculated downhole card and the geometric measurements are trustworthy today; the *labels* placed on them are not yet.